

Dimensioning

A drawing describes the shape of an object. For complete details of an object, its size description is also required. The information like distance between surfaces and edges with tolerance, location of holes, machining symbols, surface finish, type of material, quantity, etc. is indicated on the drawing by means of lines, symbols, and notes. The process of furnishing this information on a technical drawing as per a code of practice is called *dimensioning*.

4.1 ELEMENTS OF DIMENSIONING

The following are the elements of dimensioning:

1. Projection line
2. Dimension line

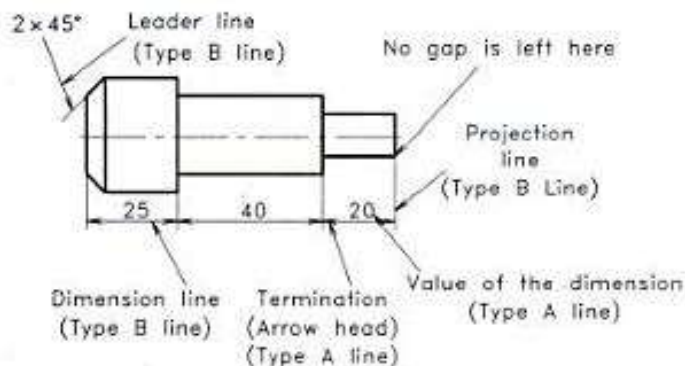


Fig. 4.1 Elements of dimensioning.

3. Leader line
4. Termination of dimension line
5. Dimensional text

These elements of dimensioning are shown in Fig. 4.1.

Projection Lines

These lines are drawn as continuous thin type B lines. They should be drawn:

1. Extending slightly beyond the respective dimension line
2. Perpendicular to the feature to be dimensioned,
3. Not crossing other lines as far as possible, and
4. May be drawn as extension of centre line or outline of the object as shown in Fig. 4.2.

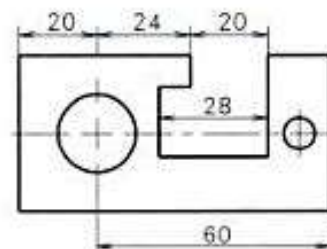


Fig. 4.2 Centre lines and outlines replace projection lines.

Dimension Lines

These lines are drawn as continuous thin type B lines. The following points may be noted while drawing a dimension line.

1. As far as possible, dimension line should not cross other lines.
2. A centre line or outlines of a part should not be used as a dimension line.
3. Dimension lines are preferred to be drawn from visible outlines and not from hidden lines.
4. A broken feature should be marked by an unbroken dimension line as shown in Fig. 4.3.



Fig. 4.3 Dimensioning of a broken feature.

Leader Lines

Leader lines are the lines referring to a feature (dimensions, object, outline, etc.) drawn as continuous thin type B lines. The tail end of the leader line should be terminated on a short horizontal bar below the lettering of a note. The head end of the leader line should be terminated in any one of the following forms (see Fig. 4.4)

1. With a dot within the outline of the object (surface).
2. With an arrow head on the outline of an object (edge).
3. Without a dot or an arrow head on a dimension line.

The following points may be noted while describing a leader line:

1. Leader line should not be parallel to adjacent dimension lines or projection lines, where confusion might arise.
2. Leader lines may be drawn at an angle not less than 30° with the horizontal or vertical.
3. The use of common leaders may be avoided.

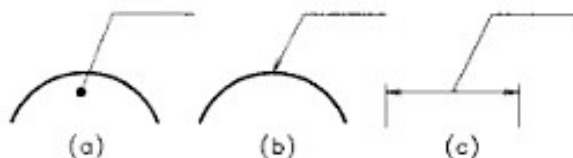


Fig. 4.4 Termination of leader lines.

Termination of Dimension Lines

Dimension lines should carry distinct terminations. Terminations are indicated either by arrowheads or by oblique strokes. The arrowheads are shown in Fig. 4.5(a).

The included angle between short lines forming the arrow head may be taken between 15° to 90° . The arrow head may be open, closed or closed and filled in. The oblique strokes are drawn as short lines inclined at 45° to the dimension line as shown in Fig. 4.5(b). While drawing arrow heads or oblique strokes the following points may be noted:

1. The size of the termination of the dimension line should be proportional to the size of the drawing on which they are used.

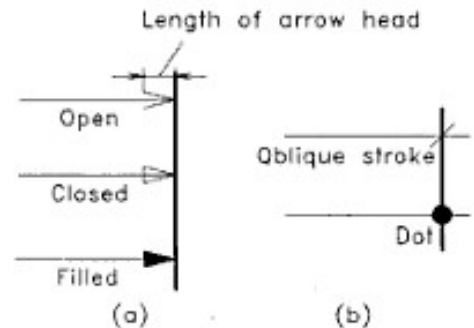


Fig. 4.5 Termination of dimension line.

2. Only one style of arrow head termination should be used on a single drawing. The shape of arrow head used in this book and that suggested for class work is shown in Fig. 4.6.

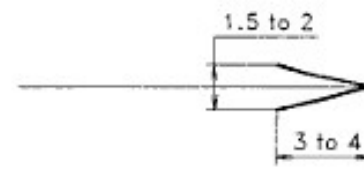


Fig. 4.6 Suggested arrow head shape.

3. Arrow heads may be shown within the limits of the dimension line, if space is available. See the left hand side dimension shown in Fig. 4.7.

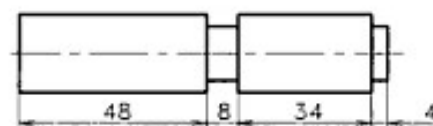


Fig. 4.7 Arrow heads within or outside the limits of dimension line.

4. Arrow head may be shown outside the intended limits of the dimension line if space is not available. See the marking of dimensional value 4 mm.

5. If the space between the projection lines is too small for an arrow head, dots or oblique strokes may be used in the place of arrow heads as shown in Fig. 4.8.
6. Only one arrow head termination is required to indicate the radius of a circle or an arc.

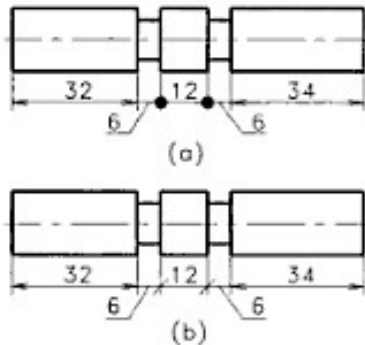


Fig. 4.8 Use of dot or oblique stroke.

Dimensional Text

Dimension is a numerical value expressed in an appropriate unit of measurement. The text size may be 3 mm to 4 mm in height depending on the drawing size. The location of the text value relative to the dimension line is decided by the *Method* of indicating them. While marking dimensional values in millimetres, there is no need of indicating mm after a value. But for the other units like cm, m or km, that should be indicated. If all the dimensions are marked in the same unit other than mm, it may be indicated as a note nearby the title block to avoid writing unit after each value.

4.2 METHODS OF INDICATING DIMENSIONAL VALUES

Two different methods of indicating dimensional values are suggested by *Bureau of Indian Standards* and are called *Method-1* and *Method-2*. Only one method is to be used in a drawing.

Method-1

In this method of dimensioning, the text should be placed aligning to the dimension line, satisfying the following conditions (see Fig. 4.9)

The dimensional values should be:

1. Placed parallel to the dimension line.
2. Placed above the dimension line.
3. Not touching the dimension line.
4. Placed at the middle of the dimension line as far as possible.

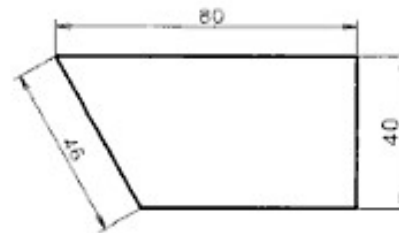


Fig. 4.9 Indicating dimensional value (Method-1).

5. Placed in such a way that it can be read either from the bottom or right hand side of the drawing.
6. Placed as indicated in Fig. 4.10 on inclined features.

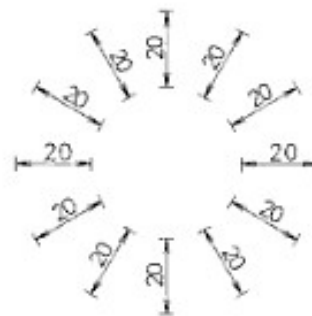


Fig. 4.10 Indicating dimensional values on oblique dimension lines (Method-1).

7. Indicated as shown either in Fig. 4.11(a) or in Fig. 4.11(b) for angular dimensioning. Here, the second one is simple, hence suggested for class work.

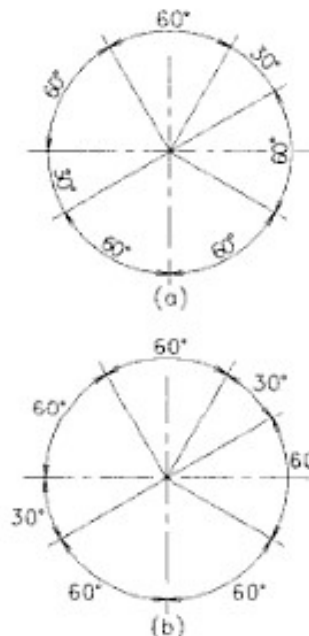


Fig. 4.11 Indicating angular dimensional values (Method-1).

Method-2

In this method of dimensioning, the text should be placed vertical, satisfying the following conditions (see Fig. 4.12).

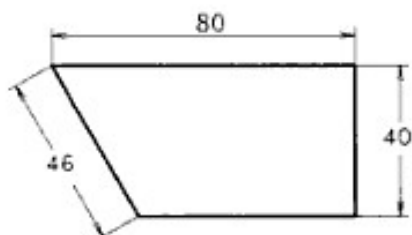


Fig. 4.12 Indicating dimensional values (Method-2).

The dimensional values should be:

1. Placed above the horizontal dimension lines and at the middle as far as possible, without interrupting the dimension line.
2. Placed at the middle by interrupting the dimension line, for non-horizontal (vertical and inclined) dimension lines.
3. Placed in such a way that it can be read from the bottom side.
4. Placed as shown on Fig. 4.13 for angular dimensioning.

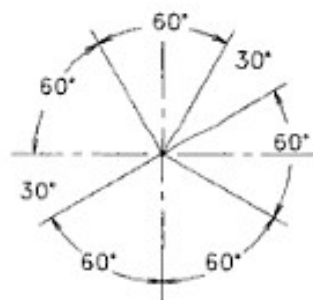


Fig. 4.13 Indicating angular dimensional values (Method-2).

4.3 ARRANGEMENT OF DIMENSION LINES

The dimensions of an object can be placed according to either *Method-1* or *Method-2*, but they are arranged in the following ways. The selection of the type of arrangement depends on the design and construction requirements.

1. *Chain dimensioning*: Here the dimension lines are drawn as a chain as shown in Fig. 4.14. This type of dimensioning is used only where the possible accumulation of tolerances does not endanger the functional requirements of the part.
2. *Parallel dimensioning*: Here the dimension lines are drawn parallel to each other as shown in

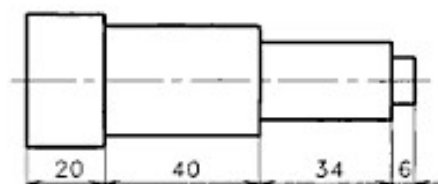


Fig. 4.14 Chain dimensioning.

Fig. 4.15. This type of dimensioning is used only where a number of dimensions of a part have a common datum feature.

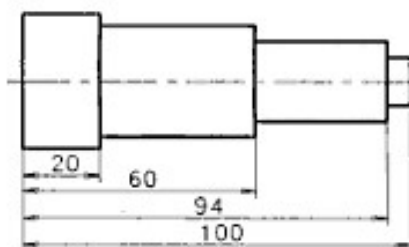


Fig. 4.15 Parallel dimensioning.

3. *Combined dimensioning*: Here a combination chain and parallel dimensioning are applied as given in Fig. 4.16, depending on the requirement.

The distance of dimension line from the object boundary or nearby dimension line should be at least 5 mm to 6 mm.

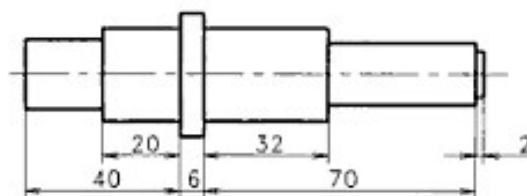


Fig. 4.16 Combined dimensioning.

4. *Superimposed running dimensioning*: This type of dimensioning is a simple parallel dimensioning and may be used where there are space limitations and where no legibility problems will arise. Here, origin is to be indicated appropriately and the opposite end of each dimension line should be terminated only with an arrow head. The dimensional values may be entered as shown in Fig. 4.17(a) or as shown in Fig. 4.17(b). This type of dimensioning can also be used to indicate dimensions in two directions as shown in Fig. 4.18.

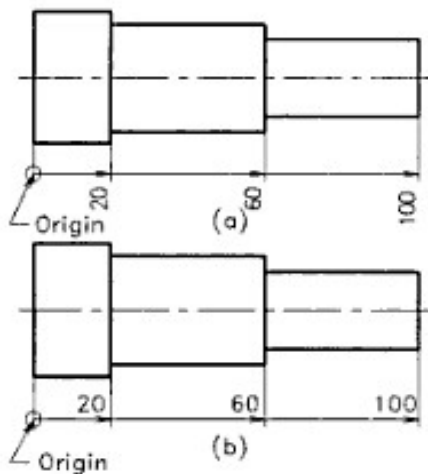


Fig.4.17 Superimposed running dimensioning in one direction.

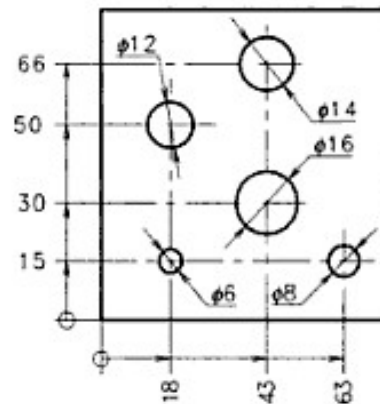


Fig.4.18 Superimposed running dimensioning in two direction.

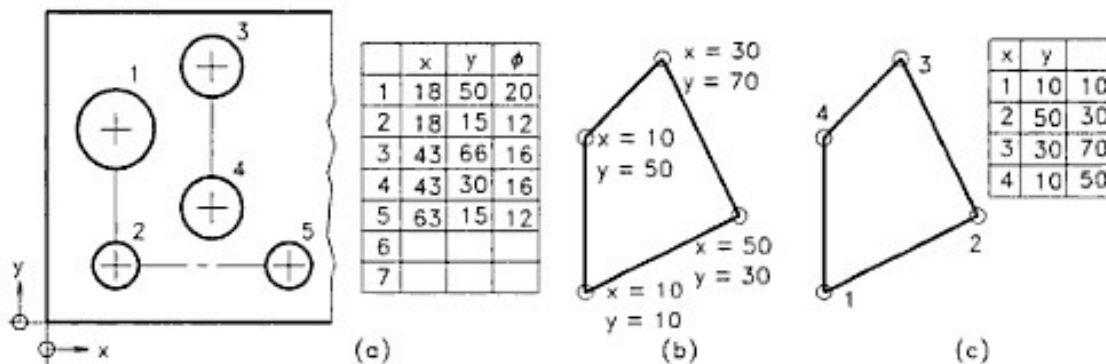


Fig. 4.19 Dimensioning by co-ordinates.

5. *Dimensioning by coordinates:* This type of dimensioning follows the principle of coordinate system of identifying the points. Figure 4.19 shows three ways of representing the dimensional values of each point under reference with respect to the origin. Here, ϕ indicates the diameter of the hole.

4.4 SHAPE INDICATION

While dimensioning an object, the shape of the feature is also indicated along with the dimension value as far as possible. This can improve the drawing interpretation. *Bureau of Indian standards* recommends the following indication (symbols) which should precede the value for dimension.

- ϕ : diameter
- R : radius
- $\square$: square.
- Sf: spherical diameter
- SR: spherical radius

Dimensioning of Geometrical Shapes

Circle

A circle shall be dimensioned by any one of the ways shown in Fig. 4.20. Symbol ϕ may be used to represent the diameter

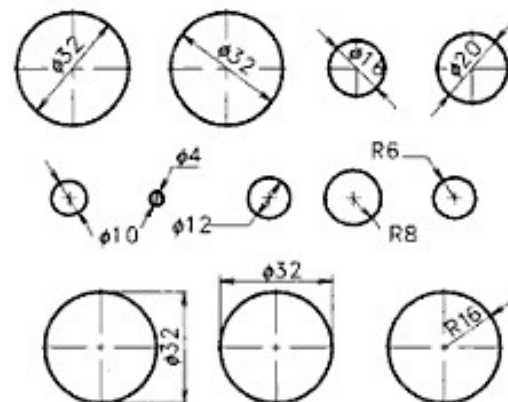


Fig. 4.20 Diameter or radius on circles.

of a circle. This symbol may be omitted in situations where the circle can be identified without confusion.

Cylindrical diameter

Diameter of a cylindrical feature shall be indicated by any one of the ways shown in Fig. 4.21. Symbol ϕ may be used to represent the diameter of a cylindrical feature. This symbol may be omitted in the situation where the cylindrical diameter can be identified without confusion.

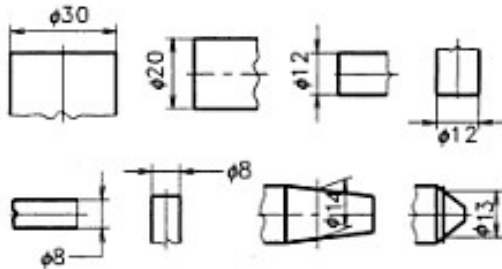


Fig. 4.21 Indicating diameter on cylindrical features.

Radius

Radius of an arc shall be dimensioned by any one of the ways shown in Fig. 4.22. Symbol R shall be used to represent the radius of an arc. Only one arrow head termination is used and it may be located depending upon the size of the feature. Generally, the arrow head is terminated on the feature outline.

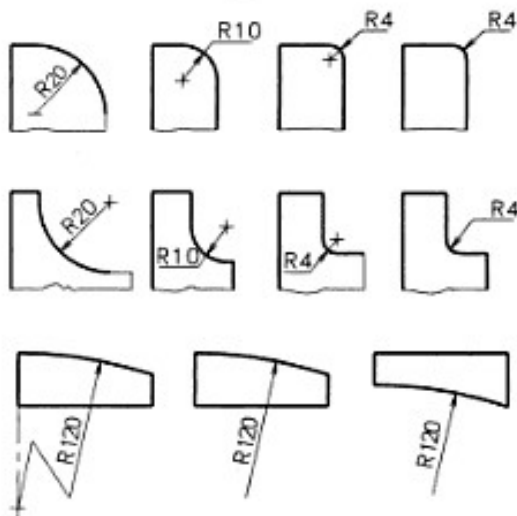


Fig. 4.22 Indicating radius on an arc.

Square

A square shall be dimensioned as shown in Fig. 4.23. Symbol $\square$ may be used to represent a square. This symbol may even be omitted in situations where the square can be identified without confusion. Square ends may be indicated by diagonals drawn as continuous thin lines.

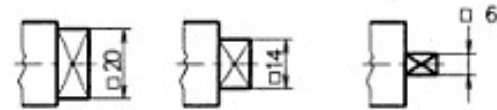


Fig. 4.23 Dimensioning on square features.

Spherical diameter and radius

Diameter and radius of a spherical feature shall be indicated as shown in Fig. 4.24. Symbol Sf shall be used to represent the spherical diameter. Here, S stands for spherical and ϕ stands for diameter. Symbol SR shall be used to represent the spherical radius. Here, R stands for radius.

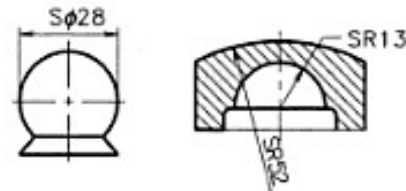


Fig. 4.24 Dimensioning on spherical features.

Chord and arc

Chord and arc shall be dimensioned as shown in Fig. 4.25.

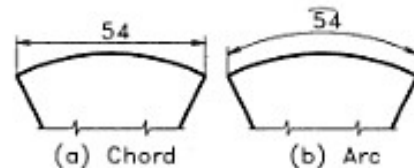


Fig. 4.25 Dimensioning on chord and arc.

Dimensioning of Miscellaneous Features

Chamfers and countersinks

External chamfers shall be dimensioned as shown in Fig. 4.26(a) and (b). Internal chamfers shall be dimensioned as shown in Fig. 4.26(c). If the chamfer angle is 45° , the indication may be simplified as shown in the figure. Countersinks are dimensioned by showing included angle with diameter or depth as given in Fig. 4.27.

4.5 SUGGESTED DIMENSIONING SYSTEM FOR CLASS WORK

Bureau of Indian Standards permits two methods of placing dimensional values (*Method-1* and *Method-2*) and various arrangements for indicating dimensions as explained above. Among them the following combination of dimensioning system is used throughout the textbook and the same is suggested for class work.

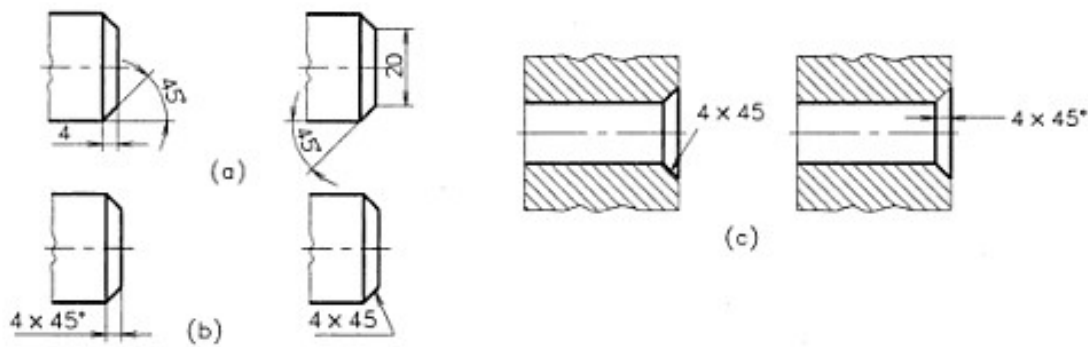


Fig. 4.26 Dimensioning on chamfers.

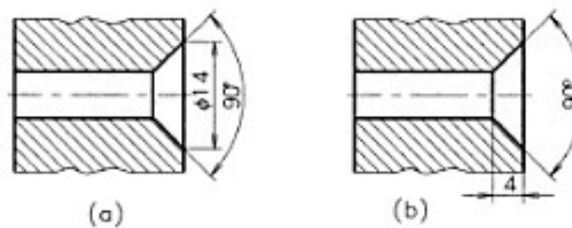


Fig. 4.27 Dimensioning on countersinks.

1. The method of placing dimensional values

The *Method-1*, of placing the dimensional value is followed throughout the textbook. This system of placing dimensional values is almost similar to the old aligned system and gives compact layout for dimensional values having more digits, tolerance, etc.

2. The arrow head

Open type arrow heads having included angle of about 15° is recommended. The length of arrow head may be 3 to 4 mm depending on the size of the pencil drawing. Use thick line (0.5 mm) for arrow head while thin line (0.25 mm) for dimension and projection lines. If the space between projection lines is too small to insert arrow heads, filled dots of about 1.5 mm is used.

3. The printing of dimensional values

The dimensional values are to be printed using letters and numerals of uniform height of about 3 to 4 mm and 0.5 mm line thickness is to be used for the purpose. They should be placed as far as possible at the middle portion, 0.5 to 1 mm above the dimension line. HB pencil is recommended for printing text using stencil.

4. Units for dimensioning

Millimetre is the unit recommended by BIS for all engineering drawings, except for special requirements. Hence, any value written as dimension will be normally

considered as in millimetres. Therefore it is not allowed to write "mm" after a dimensional value and also is not necessary to write *All dimensions are in mm* nearby the title block. But, if any unit other than mm is used, then that should be noted either after the dimensional value (as cm, m, etc.) or as a general instruction.

5. Scale of drawing

Generally class work drawings are prepared in 1:1 scale. If a different scale is used, it should be noted below the drawing and in the space provided inside the title block. At the same time, whatever may be the scale of drawing, the actual given dimensions should be written on the dimension lines. For example, 50 m length, drawn to a scale as 100 mm, should be dimensioned as 50 m itself.

6. Answers and captions

Captions like the name of object, views etc., should be written below the drawing using capitals only, of 3 to 5 mm letters. The question number is written at the left top side of the figure and may be enclosed by a circle of 10 mm diameter to identify it.

4.6 RULES OF DIMENSIONING

1. Dimension lines, projection lines, extension lines, and leader lines are drawn as continuous thin lines.

2. Projection lines should extend slightly beyond the dimension line (1 to 2 mm).
3. Extension and dimensioning lines should not cross other lines unless that is unavoidable.
4. Extension lines are drawn in a direction perpendicular to the feature to be dimensioned. They may be drawn obliquely but parallel, to each other on special requirements.
5. Dimensions are to be given from visible outlines rather than from hidden lines.
6. Each end of the dimension line should be defined by an arrow head.
7. Adjacent arrows may be replaced by a clearly marked dot or slash.
8. If dimensioning inside a hatched portion is unavoidable, the hatching lines should not cross the dimensional text.
9. Two systems of dimensioning should not be mixed up in one drawing. Only one system of dimensioning

(Method 1 or 2) should be followed throughout a particular drawing.

Example 4.1

Dimensioned view of an L shaped plate is shown in Fig. 4.28(a). Copy the view and place the dimensions as per parallel dimensioning of Method-1, taking the left hand bottom corner as the reference point.

Refer to Fig. 4.28(b).

1. Draw the given L shaped plate.
2. Place the dimensions correctly, following the Method-1 and parallel dimensioning taking the left hand bottom corner as the reference point.

Example 4.2

Read the dimensioned drawing of a lock plate shown in Fig. 4.29(a). Redraw the figure to full size and dimension it following Method-1 of chain dimensioning as per Bureau of Indian Standards.

Refer to Fig. 4.29(b).

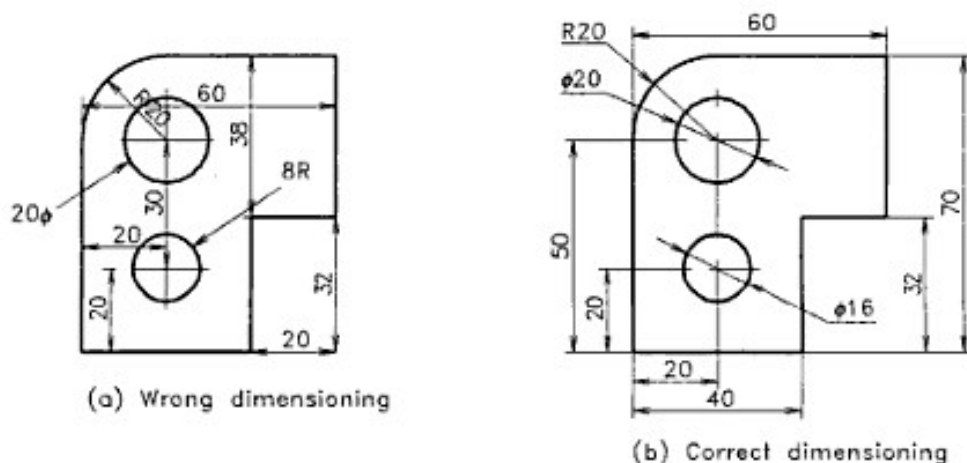


Fig. 4.28 L-plate (parallel dimensioning).

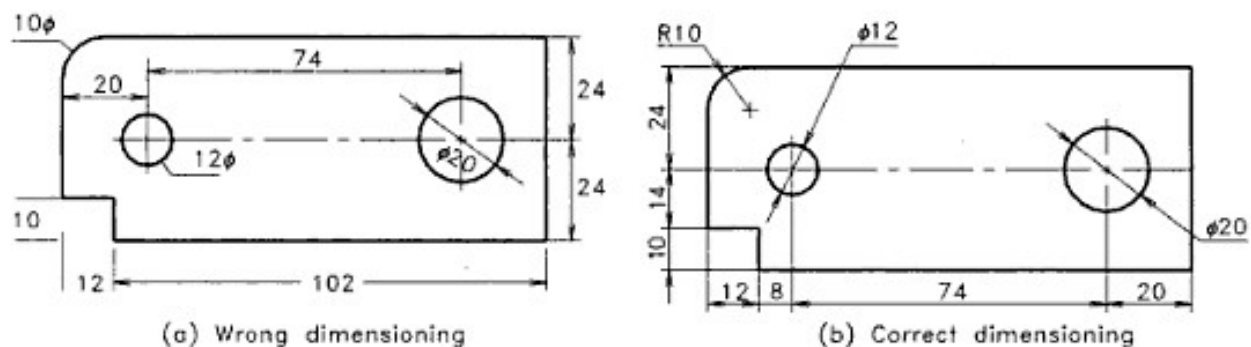


Fig. 4.29 Lock plate (chain dimensioning).

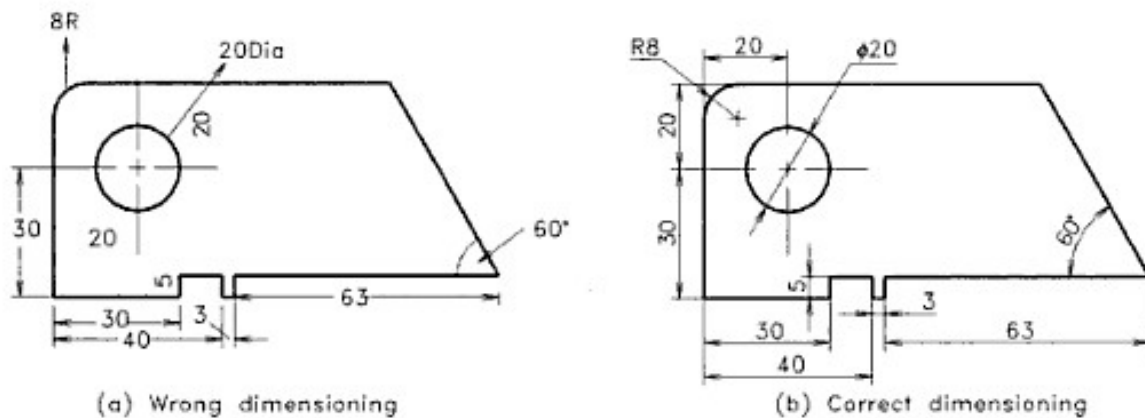


Fig. 4.30 A template (combined dimensioning).

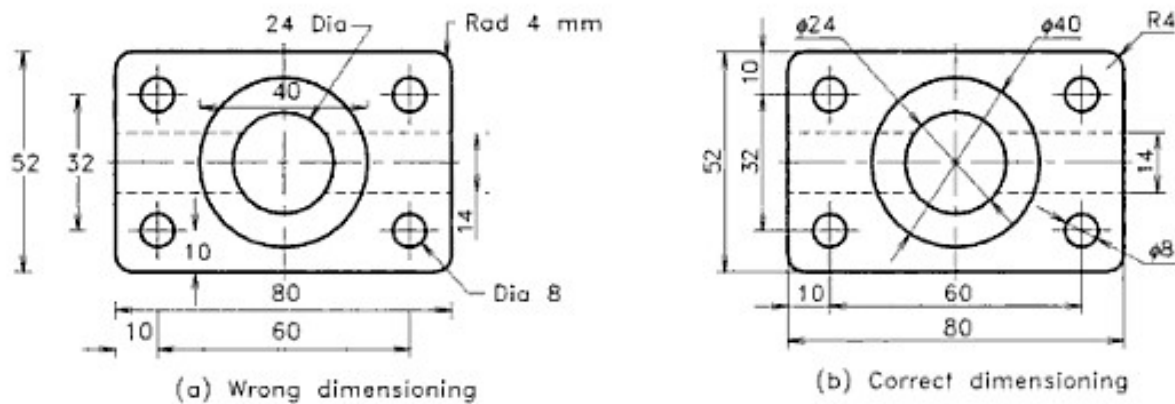


Fig. 4.31 A rod support (combined dimensioning).

Example 4.3

Figure 4.30(a) gives a template dimensioned wrongly. Copy the figure to scale 1:1 and place the dimensional values as per Method-1 and combined dimensioning. Refer to Fig. 4.30(b).

Example 4.4

Copy the drawing of a rod support given in Fig. 4.31(a) and enter the dimensions following Method-1 of combined system of dimensioning.

Refer to Fig. 4.31(b).

Example 4.5

Figure 4.32(a) shows a wrongly dimensioned gland plate. Redraw the figure and dimension it as per Method-1 of BIS.

Refer to Fig. 4.32(b).

Example 4.6

Draw the given Fig. 4.33(a) to a scale 2:1 after taking measurements from the figure and dimension them following combined dimensioning system of Method-2. The measurements are to be corrected to 1 mm accuracy.

Refer to Fig. 4.33(b).

1. Measure the dimensions of the figure and draw to the double size of each dimension.
2. Mark the dimensional values as per Method-2.

Example 4.7

Copy the cylindrical machine part given in Fig. 4.34(i) using 1:1 scale and place dimensional values changing from Method-1 to Method-2.

Refer to Fig. 4.34(b).

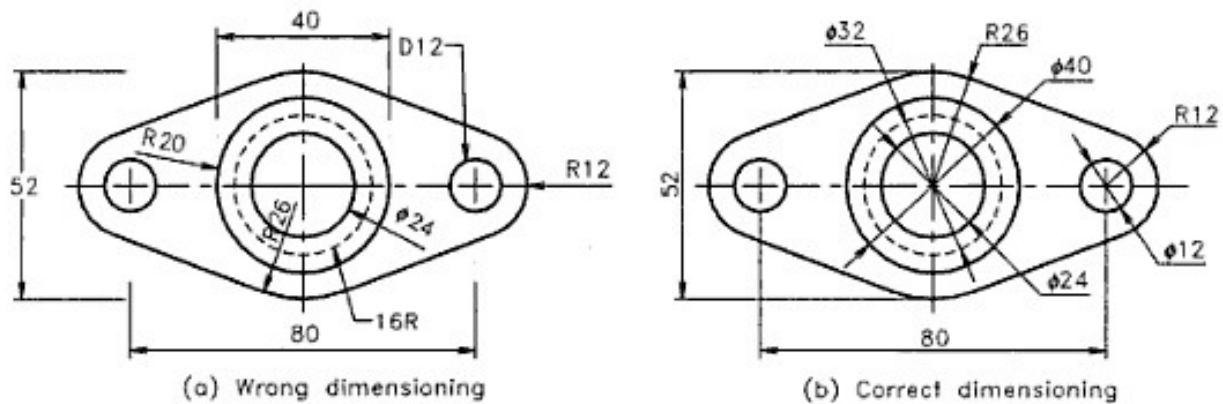


Fig. 4.32 A gland (Method-2 dimensioning).

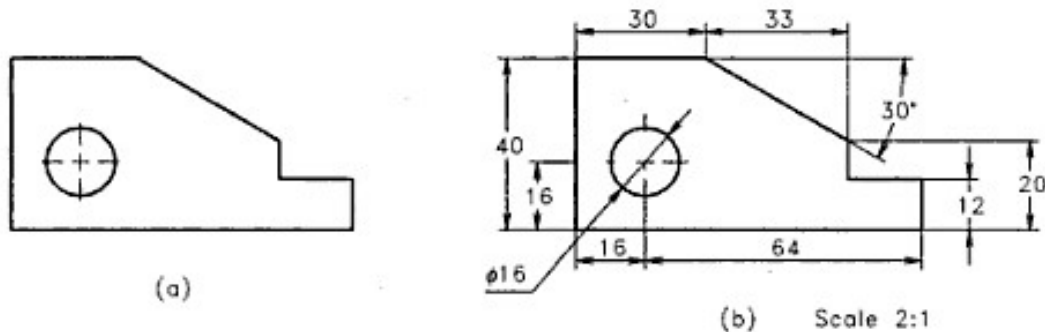


Fig. 4.33 Wedge plate (Method-2 dimensioning).

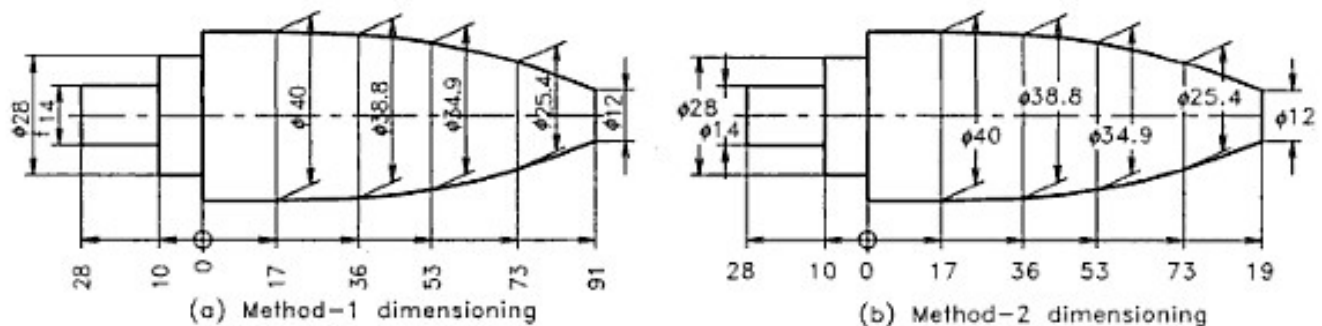


Fig. 4.34 A cylindrical machine part.

EXERCISES

Method-1 dimensioning

- The following figures give drawings of flat engineering objects with wrong dimensioning. Copy them and insert the dimensional values as per combined dimensioning of Method-1.

(a) Guiding plate (Fig. 4.35)

- Lock washer (Fig. 4.36)
- U-plate washer (Fig. 4.37)
- Template (Fig. 4.38)
- Lock washer (Fig. 4.39)
- Angle plate (Fig. 4.40)

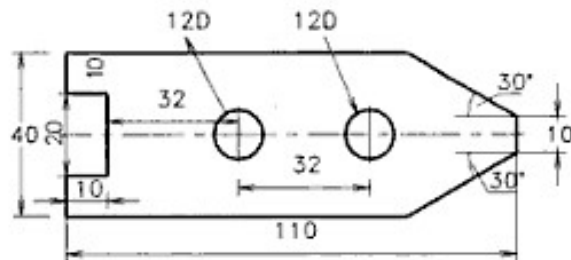


Fig. 4.35 A guiding plate (Wrong dimensioning).

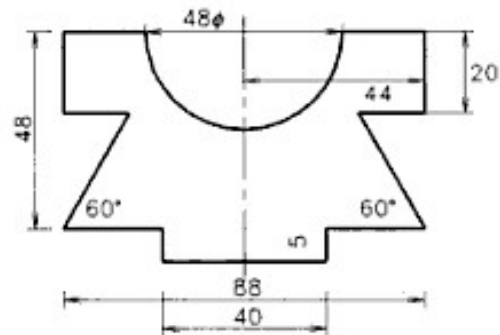


Fig. 4.38 A template (Wrong dimensioning).

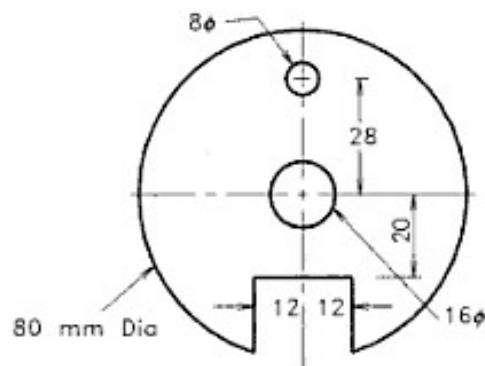


Fig. 4.36 A lock washer (Wrong dimensioning).

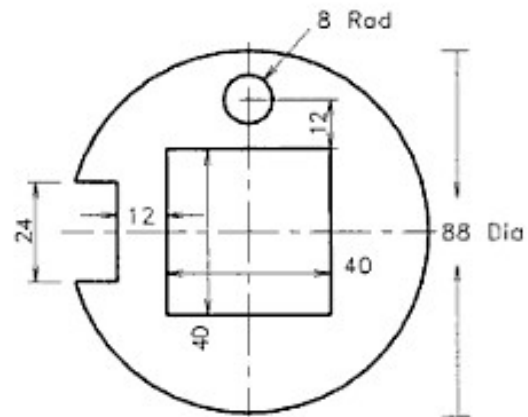


Fig. 4.39 A lock washer (Wrong dimensioning).

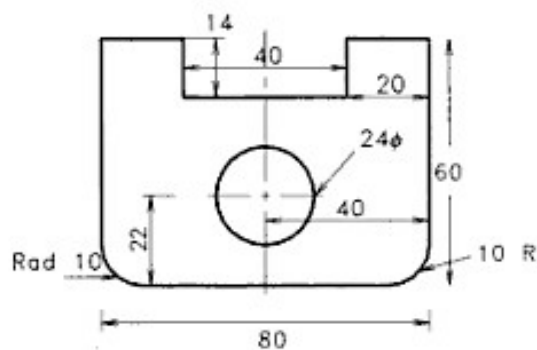


Fig. 4.37 A U-plate washer (Wrong dimensioning).

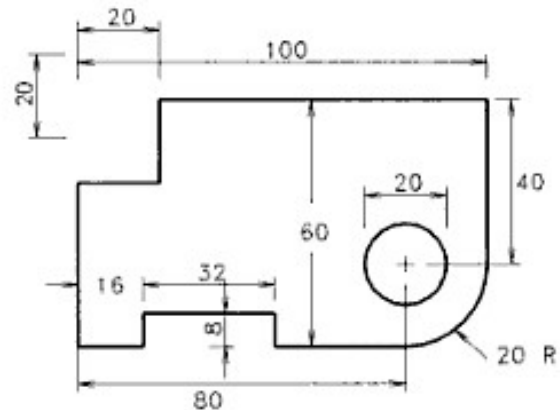


Fig. 4.40 An angle plate (Wrong dimensioning).

Method-2 dimensioning

2. The following figures give drawings of objects with wrong dimensioning. Copy them and insert the dimensional values as per combined dimensioning of Method-2.

- Lock washer (Fig. 4.39)
- Angle plate (Fig. 4.40)
- Go-no-go gauge (Fig. 4.41)
- A machine part (Fig. 4.42)

- Figure 4.43 shows drawing of an axle with cylindrical shape. Draw the figure to a scale 2:1 after taking measurements from it and dimension them following Method-2 of BIS. The measurements are to be corrected to 1 mm accuracy.
- Copy the drawing of a machine part given in Fig. 4.44 and enter the dimensions following Method-2 of combined system of dimensioning.

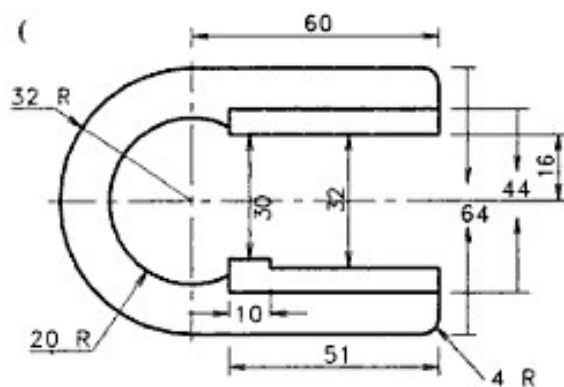


Fig. 4.41 Go-no-go gauge (Wrong dimensioning).

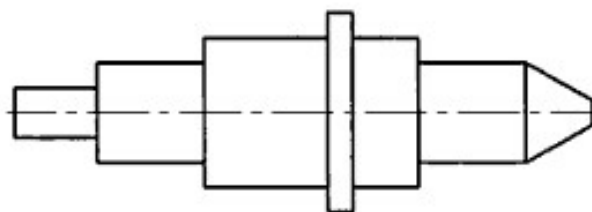


Fig. 4.43 Axle.

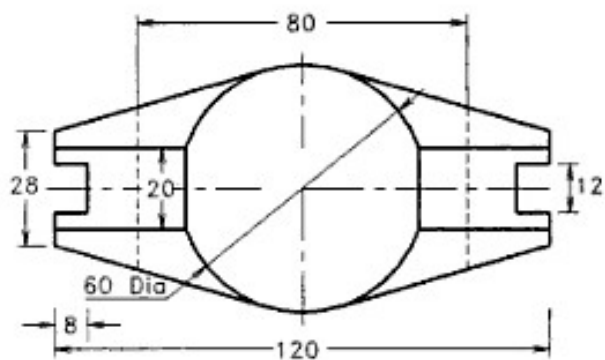


Fig. 4.42 A machine part (Wrong dimensioning).

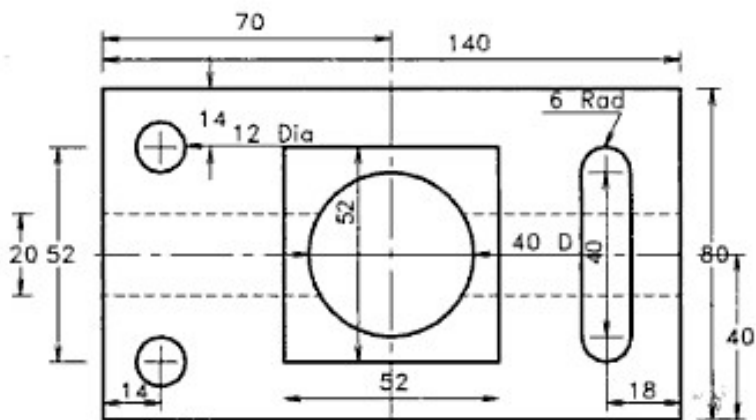


Fig. 4.44 A machine part (Wrong dimensioning).